

Code-Layout, Readability and Reusability

Date	01 July 2026
Team ID	ghousiyaunnissa@gmail.com
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Follows standard PEP 8 4-space indentation across app.py and scripts.
2	Proper File Structure	Files and folders are logically organized	Yes	Clean layout separating templates/, static/, and core predictive files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables match descriptive financial parameters (e.g., total_annual_income).
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict_approval() and load_model() explain their mechanics.
5	Code Comments	Inline and block comments explain logic	Yes	Comprehensive block descriptions cover preprocessing array steps.
6	Modular Design	Code is split into reusable functions/modules	Yes	Data scaling, template encoding, and prediction run on decoupled logic.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Consolidated training runs and feature setups remove code clones.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Robust try-except structures capture malformed input data payloads securely.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low) High
1	approval_model.pkl	Python / Scikit-Learn Classifier System	Flask production routing pipeline	
2	model_columns.pkl	Python / Pickle Blueprint Map	Categorical encoding alignment scripts	High
3	pd.get_dummies Feature Transformation Block	Python / Pandas Data Matrix Tool	Input preprocessing functions	High
4	Form Validation Middleware	Python / Flask App Microframework	Inbound request sanitization modules	Medium
5	CSS Grid Core Styling Layout	HTML5 / CSS3 Responsive Grid Design	Application front-facing portals	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Highly structured standard workspace environment directory structure separating application logic from serialized matrix stores.
Readability	5	Highly clean script variables and parameters matching actual financial metric types perfectly.
Reusability	5	Serialized byte streams and data transformation pipeline structures make portability effortless.
Documentation / Comments	5	Block headers explain the code layout structure and exception traps thoroughly.
Overall Score	5 / 5	Excellent execution maintaining top-tier software quality assurance guidelines.